

SQL INDUSTRIAL PRACTICAL TASK

Employee Management System - DDL + Basic DML

60 Small Practical Tasks - Basic to Intermediate

Student Name _____

Mobile Number _____

Batch _____ Date _____

Objective: Practice only core SQL DDL and basic DML operations using one simple employee-management database. No joins, ORDER BY, GROUP BY or advanced reporting are required.

Suggested target for 1 hour: Minimum Task 1-40, Good Task 1-50, Excellent all 60.

Part A - Database & Table Creation

Task 1: Create a database named **company_db**.

Task 2: Use/select the **company_db** database.

Task 3: Create a table named **employees** with the following columns:

Task 4: Display the structure of the **employees** table.

Task 5: Display all tables available inside **company_db**.

employees table structure:

employee_id	INT PRIMARY KEY
employee_name	VARCHAR(100)
department	VARCHAR(50)
salary	DECIMAL(10,2)
city	VARCHAR(50)
joining_date	DATE
status	VARCHAR(20)

Part B - INSERT Operations

Task 6: Insert the following employee record:

101, Rahul Patil, Development, 45000, Pune, 2026-01-10, Active

Task 7: Insert the following employee record:

102, Priya Sharma, Testing, 38000, Mumbai, 2026-02-15, Active

Task 8: Insert the following employee record:

103, Amit Joshi, Development, 52000, Pune, 2025-12-05, Active

Task 9: Insert the following employee record:

104, Sneha Kulkarni, HR, 35000, Nashik, 2026-03-20, Active

Task 10: Insert the following employee record:

105, Rohan Deshmukh, Support, 30000, Mumbai, 2026-04-01, Inactive

Task 11: Insert the following employee record:

106, Anjali More, Testing, 42000, Pune, 2026-05-12, Active

Part C - Basic SELECT Queries

- Task 12: Display all records from employees.
- Task 13: Display only employee_name.
- Task 14: Display employee_name and salary.
- Task 15: Display employee_name, department, and city.
- Task 16: Display employees from Pune.
- Task 17: Display employees from Mumbai.
- Task 18: Display Development department employees.
- Task 19: Display Testing department employees.
- Task 20: Display only Active employees.
- Task 21: Display only Inactive employees.
- Task 22: Display the employee whose employee_id = 103.
- Task 23: Display the employee whose name is Priya Sharma.
- Task 24: Display employees whose salary is greater than 40000.
- Task 25: Display employees whose salary is less than 40000.
- Task 26: Display employees whose salary is equal to 35000.
- Task 27: Display employees whose salary is greater than or equal to 42000.
- Task 28: Display Pune employees whose status is Active.
- Task 29: Display Development employees whose salary is greater than 45000.
- Task 30: Display employees from Pune or Mumbai.

Part D - UPDATE Operations

- Task 31: Update Rahul Patil's salary from 45000 to 48000.
- Task 32: Update Rohan Deshmukh's status from Inactive to Active.
- Task 33: Update Sneha Kulkarni's city from Nashik to Pune.
- Task 34: Update Priya Sharma's department from Testing to Development.
- Task 35: Update employee 106 salary to 45000.
- Task 36: Increase salary of employee 103 by 3000.
- Task 37: Increase salary of all Testing employees by 2000.
- Task 38: Change all employees from Mumbai to Mumbai Branch in the city column.

Part E - DELETE Operations

- Task 39: Delete the employee with employee_id = 105.
- Task 40: Delete the employee named Rohan Deshmukh.
- Task 41: Delete all employees whose status is Inactive.
- Task 42: Delete employees whose salary is less than 30000.
- Task 43: Delete the employee with employee_id = 104.

Part F - ALTER TABLE / DDL Practice

Task 44: Add a new column: email VARCHAR(100).

Task 45: Add a new column: mobile VARCHAR(15).

Task 46: Change the size of city to VARCHAR(100).

Task 47: Rename column employee_name to name.

Task 48: Delete the mobile column.

Task 49: Add a column: experience INT.

Task 50: Update experience for one employee.

Part G - Table-Level DDL Practice

Task 51: Create another table named departments with the structure shown below.

department_id	INT PRIMARY KEY
department_name	VARCHAR(100)
location	VARCHAR(100)

Task 52: Insert 3 department records.

Task 53: Display all department records.

Task 54: Update one department location.

Task 55: Delete one department record.

Task 56: Rename departments table to company_departments.

Task 57: Display the structure of company_departments.

Task 58: Remove all records from company_departments while keeping the table.

Task 59: Delete the company_departments table completely.

Task 60: Display all remaining tables in the database.

Topics Used in This Practical

CREATE DATABASE
USE
CREATE TABLE
DESC / DESCRIBE
SHOW TABLES

INSERT
SELECT
WHERE
AND
OR

UPDATE
DELETE

ALTER TABLE
ADD COLUMN
MODIFY COLUMN
RENAME COLUMN
DROP COLUMN
RENAME TABLE
TRUNCATE TABLE
DROP TABLE

Reminder: To display all remaining tables in the selected database, use SHOW TABLES;